Tidal Hiccups and Oxic Sighs : Machine Learning Meets Marshland Respiration

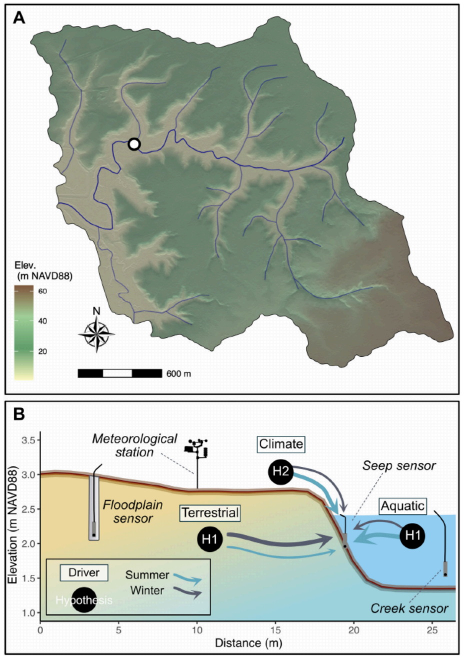
Formalizing Classification Methodologies for Environmental Drivers of Dissolved Oxygen

Tanmay Pani · Eric Britt

The data for this study is the courtesy of Dr. Ruby Ghosh and Opti-O2

Opti O2: Coastal Wetland DO_2 Dynamics

- **The Site**: Beaver Creek, WA – a former freshwater floodplain transitioning to a coastal wetland.
- **The Data**: 6-year, 5-minute resolution time-series of subsurface **Dissolved Oxygen (DO_2)**, **Hydrology** (*Water level*, *Salinity*, etc.), and **Weather** (*Temperature*, *Precipitation*, etc.) timeseries data
- **The Phenomena**:
  - **DO_2 ~ 0** for 92% of timestamps (Anoxic baseline)
  - **DO_2 event:** Sufficient departure from the baseline ($\Delta DO_2 > 0.1$mg/L)
- **The Challenge**: Classifying DO_2 events into:
  - **Hot Moments** (abrupt asymmetric and tidal/salinity-driven)
  - **Oxic Pulses** (symmetric and freshwater/precipitation-driven)



Event Detection

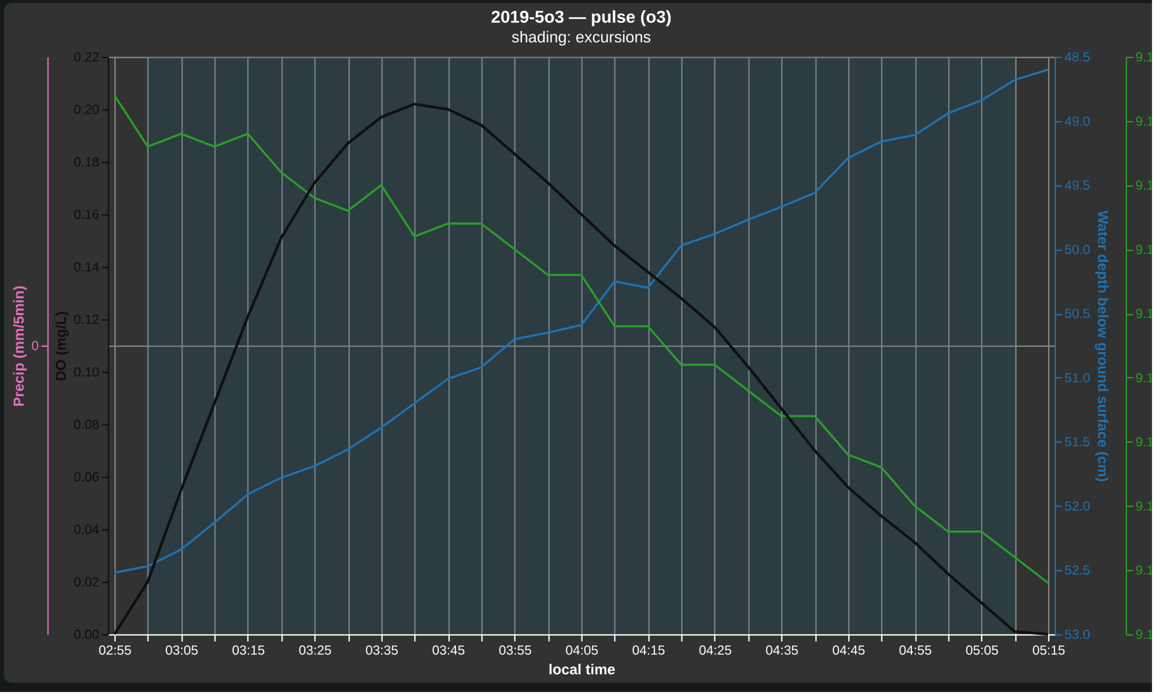
- **Moments (expert-labeled):** Expert detected events based on hydrology (DO_2, flooding, tides, precipitation, etc.) and each event split into distinct hot moments (tidal) and oxic pulses (freshwater).
- **Excursions (auto-detected):** Sustained DO_2 departure from anoxic baseline $DO_2 = 0$ mg/L.

Event 2019-5o3 — pulse (o3)

Shade excursions moments

2019-5o3 — **pulse** (o3)

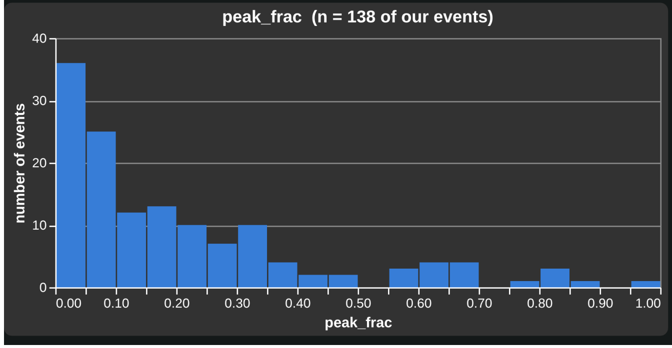
- window: **2019-07-02 02:55** → **2019-07-02 05:15** (2.3 h)
- classification code **o3**: oxic pulse — prior to flooding
- peak DO_2: **0.20 mg/L**
- salinity step: **+2.44**
- water level step: **-8.44**
- antecedent precipitation (24 hours): **0.0**
- flooding: **True**
- number of DO excursions: **1 event(s)**

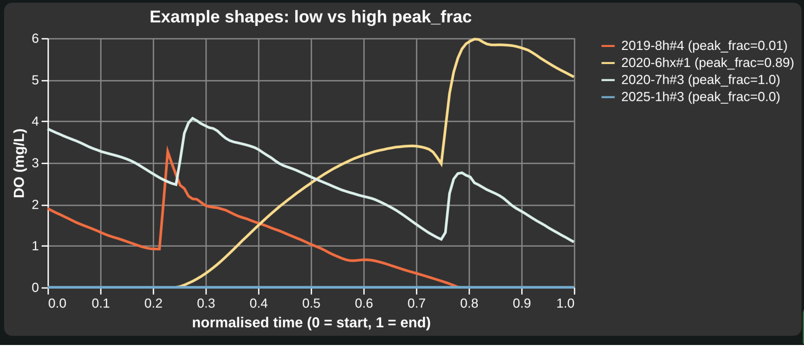


Engineered Features: Examples

Time-To-Peak | Rate_{DO2} | ΔSalinity | ΔT

Time-To-Peak (peak_frac) = $\frac{\arg\max_i d_i}{n-1}$



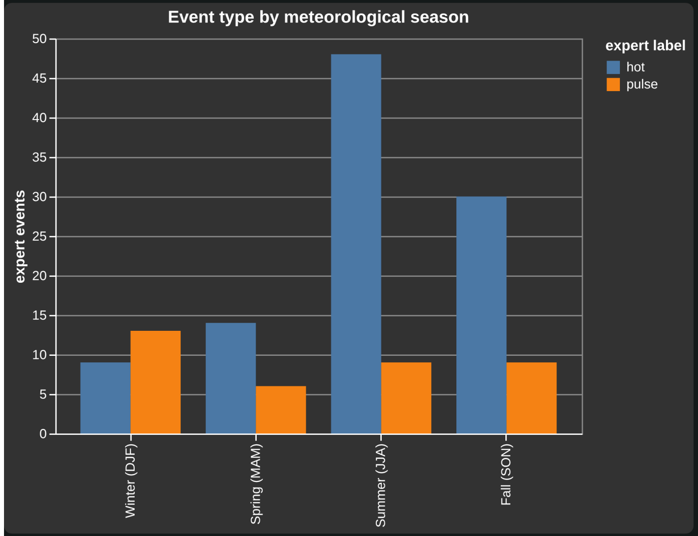


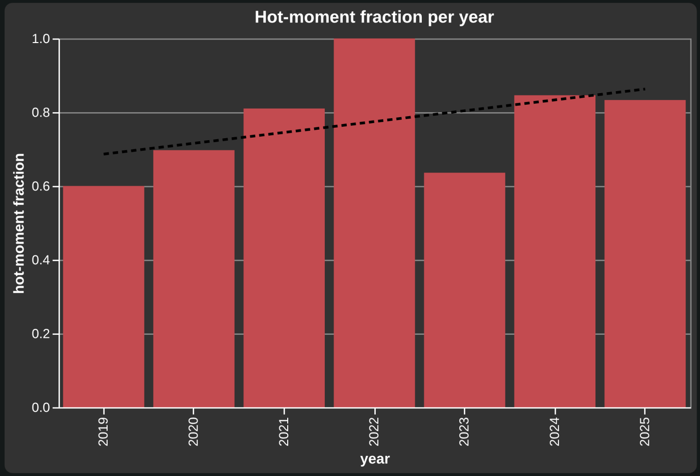
made v

Modelling

- **Engineered Features**:
    - Evaluates tabular models on 24 domain-engineered features
    - Calculated based on shape of oxic event (abrupt vs slow), hydrology and antecedent precipitation
    - **Models**: Linear (*Logistic Regression*), Tree-based (*XGBoost, CatBoost*), and Deep/Foundation (*TabPFN*).
- **Raw Sequences / Time Series**:
    - Evaluates deep learning models directly on the raw 5-minute multivariate time-series waveforms.
    - **Models**: Sequences transformed by ROCKET, passed on to XGB/CatBoost/Logistic/TabPFN, InceptionTime-Lite (CNN).
- **Cross Validation**: Evaluated via 5x5 StratifiedGroupKFold and Leave-One-Group-Out (LOGO) cross-validation, holding out expert events from training
- **Temporal Generalization**: Includes a 70/30 chronological split to test model robustness against multi-year ecosystem salinization and drift.

Label Imbalance

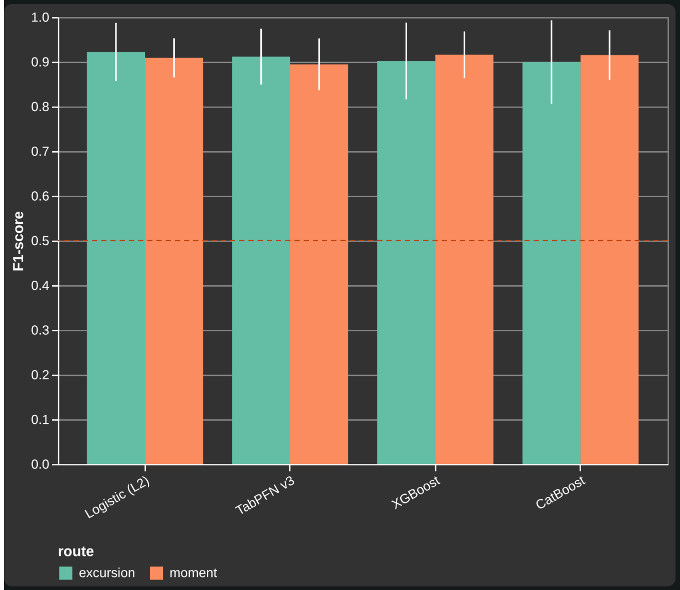




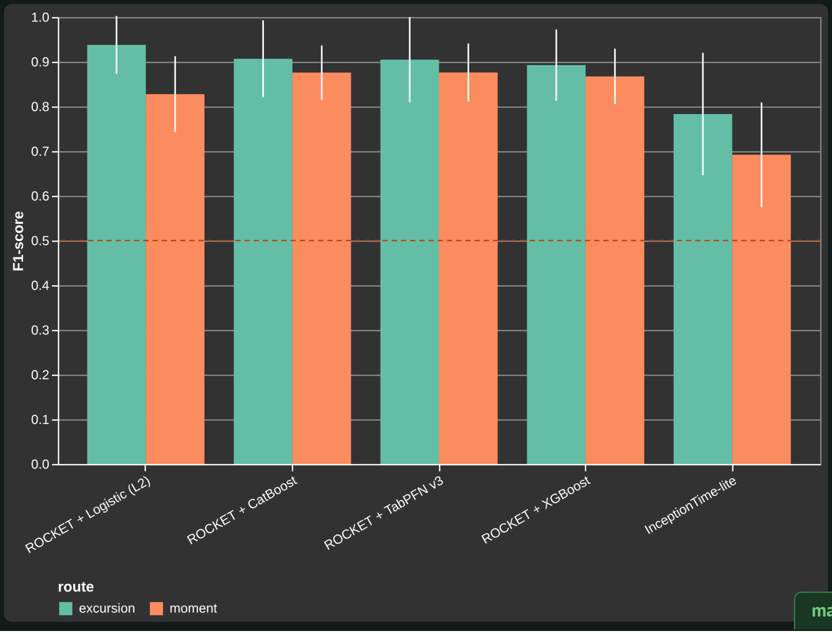
- Label
has se
year-b
- Hot m
domir
summ
freshv
incurs
- Oxic p
domir
(less s
incurs
freshv
- Hot m
freque
with y
rise?)

Model comparison — Macro-F1 (grouped 5×5 CV)

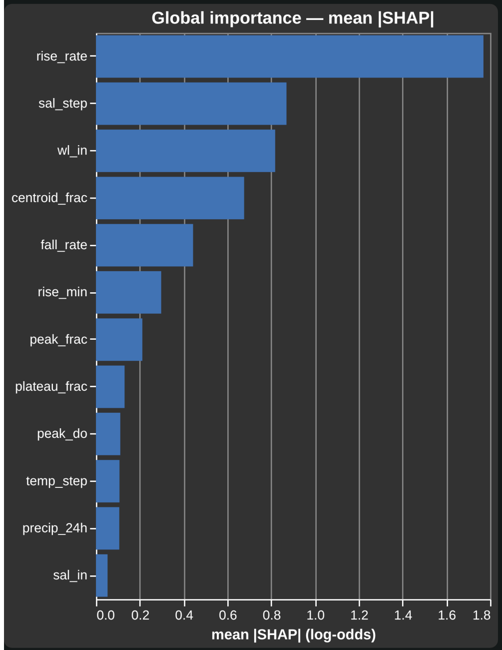
Engineered features (excursion vs moment):

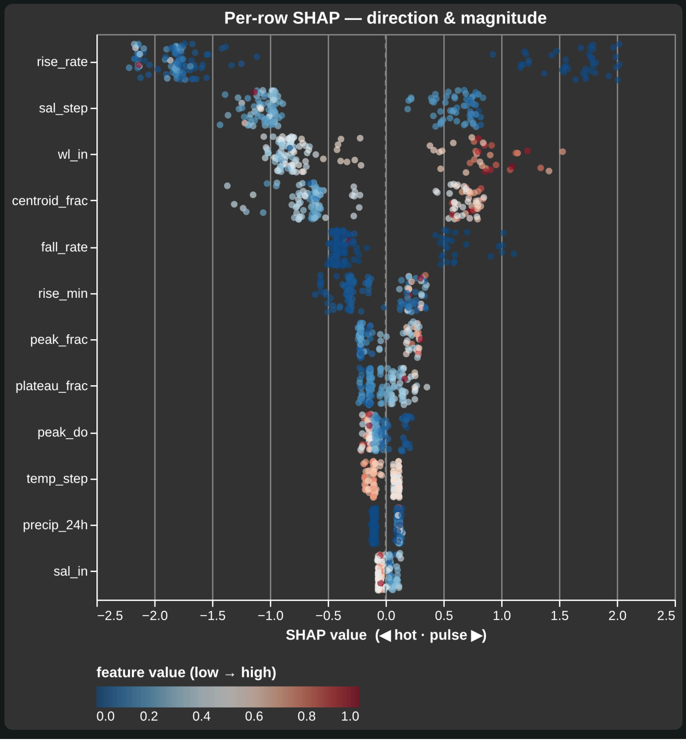


ROCKET / InceptionTime on raw curves:

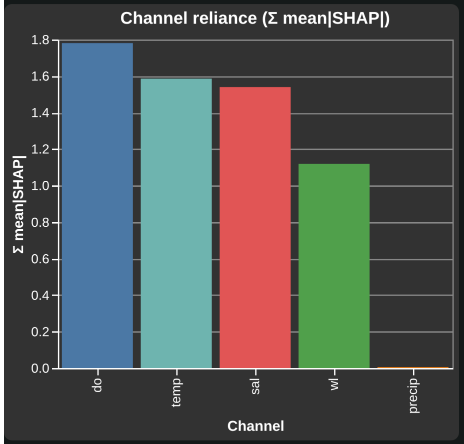


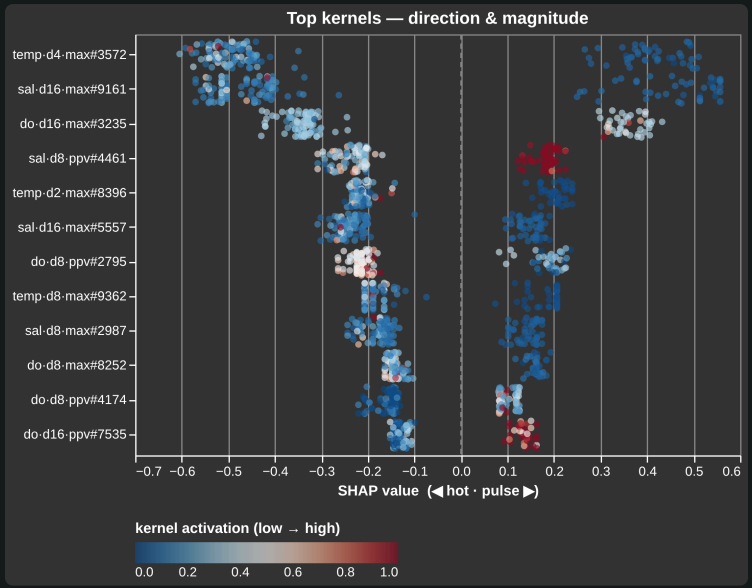
Which drivers carry hot-vs-oxic signal? — XGBoost (engineered)

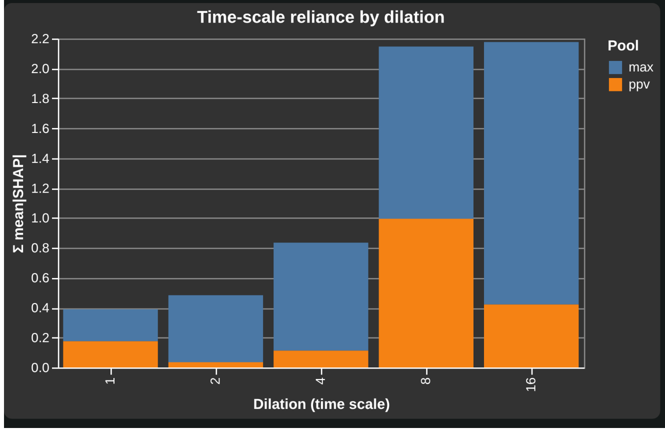




Which drivers carry hot-vs-oxic signal? — ROCKET + XGBoost (raw curves)



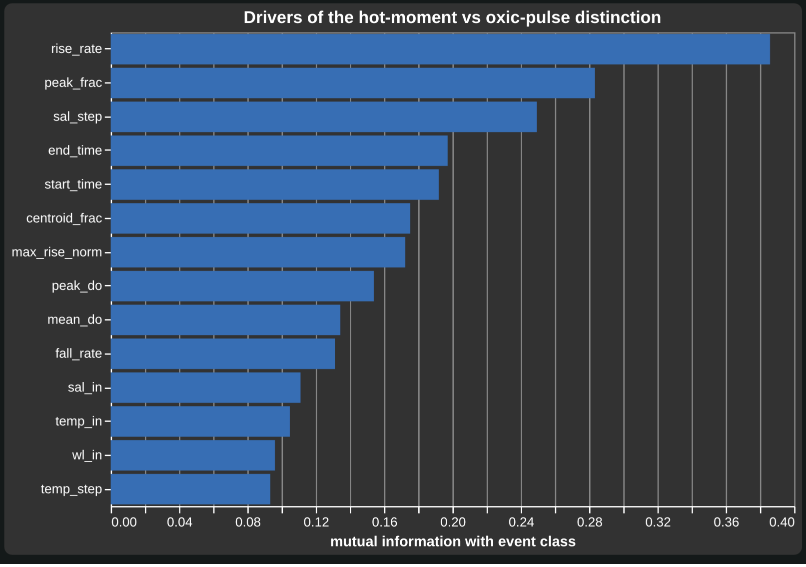




Which drivers carry hot-vs-oxic signal? — read-out

- **SHAP, Engineered Features (XGBoost, Logistic, TabPFN):**
    - Best → **Hydrology-based** e.g., $\frac{d(DO2)}{dt}$ (*rise_rate*), $\frac{t_{\text{peak}-\text{DO}_2} - t_{\text{start}}}{t_{\text{end}} - t_{\text{start}}}$ (*peak_frac*), $\Delta\text{Salinity}$ (*sal_step*), $\max\,(DO_2)$ (*peak_do*)
    - Worst → **Weather-based:** Air Temperature (temp_in), Antecedent Precipitation (precip_168h)
- **SHAP, Time Series (ROCKET+XGBoost, InceptionTime):**
    - DO_2 and Temperature most important, Precipitation least
    - ROCKET kernels with bigger dilation more important (~40-80 min correlations)
- Mutual information (unsupervised, right) validates conclusions from SHAP →

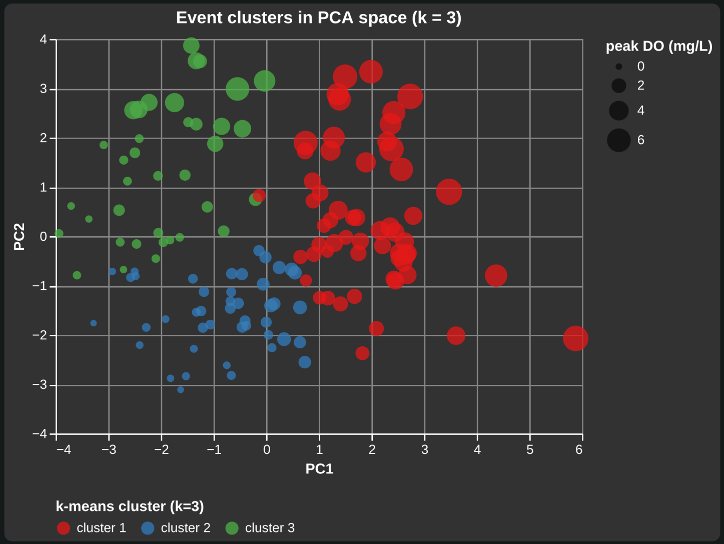
Unsupervised (mutual info)

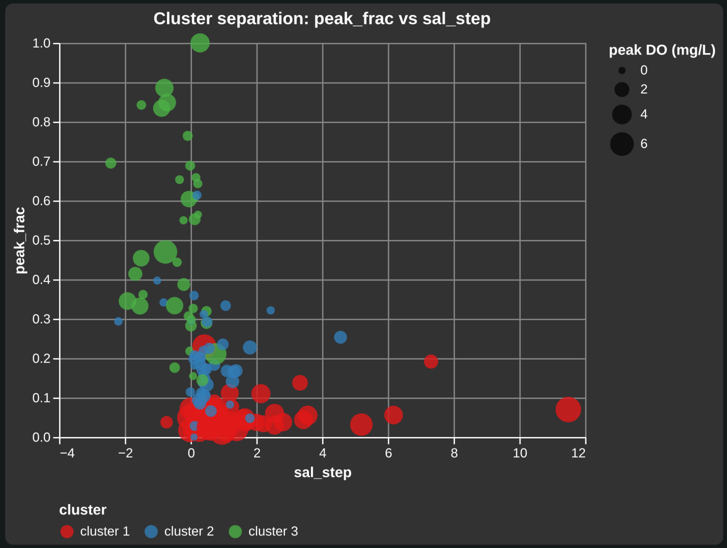


Unsupervised: Clustering

k (number of clusters) 3

x axis sal_step y axis peak_frac





cluster str	hot u32	oxic u32	total u32	purity f64
cluster 1	57	0	57	1.000
cluster 2	37	7	44	0.841
cluster 3	7	30	37	0.811

3 rows, 5 columns *No selection*

Weak-supervision labels — physics-only vs expert

- **Labeling functions** vote *hot* / *pulse* / *abstain* using single feature,
- Accuracy-weighted vote aggregation → *weak-supervised labels*
- Taxonomy (Approx.):
  1. Abrupt rise + Slow fall + coincident salinity / water-level step → **hot moment**
  2. Symmetric curve + Antecedent precipitation → **oxic pulse**

Conclusions and Future work

- Formalized classification of Hot moments and Oxic pulses using timeseries data from Beaver Creek, WA
- Developed various Deep-Learning (DL) classifiers trained engineered features, encoded features and raw time series
- All model validated using various cross validation methods and non-DL models (XGBoost, Logistic)
- Validated the expert labels against our own labelling algorithm
- Feature importances attributed using SHAP (supervised) and validated against mutual information (unsupervised)
- Groundwork laid for unsupervised classification (KMeans, Weak-Supervision Label)